
ME 206: Mechanics of Solids

Course Code	ME 206
Course Name	Mechanics of Solids
L-T-P-C	2.5-0.5-0-3
Prerequisites	Consent of the faculty member
Intended for	B.Tech.
Distribution	Core
Approval	8th Senate (Oct 2015)

Course Contents

- Free body diagram, Conditions for equilibrium: statically determinate & indeterminate
- Mechanics of small deformation: Geometric compatibility & force deformation law (for uniaxial loads)
- Special kinds of load: Transverse loaded slender member : Shear force & Bending moment
- Stress and Strain: Proper definition of stress and strain
- Theory of yielding
- Shaft : Circular cross-section shaft under uniform & varying load (torque), Twisting deformation of shaft
- Bending Stresses
- Deflection of Beam (superposition theorem), Castigliani-II theorem: Energy method (unit load method)
- Buckling of column (Brief discussion in the context of elastic instability)

Text and Reference Books

1. Timoshenko S. P., and Gere J. M., **Mechanics of Materials**, 2nd Edition, CBS Publishers, 2002.
2. Crandall S. H., Dahl N. C., and Lardner T. J., **An Introduction to the Mechanics of Solids**, 2nd Edition, McGraw-Hill, 1999.
3. Hearn E. J., **Mechanics of Materials**, 3rd Edition, Pergamon, 2003.
4. Higdon A., Ohlsen E. H., Stiles W. B., Weese J. A., and Riley W. F., **Mechanics of Materials**, John Wiley & Sons, 1989
5. Popov E. P., Nagarajan S., and Lu Z. A., **Mechanics of Materials**, 2nd Edition, Prentice-Hall of India, 2002.
6. Johan K. C., **Text Book of Machine Drawing**, PHI Learning Pvt, 2009.
7. SP 46: 1988 Engineering Drawing Practice for Schools and Colleges, Bureau of Indian Standards, 1988.